

Project Presentation

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Background

- Dataset: Research Paper NIPS2022
 - comprises a total of 6 different synthetic bank account fraud tabular datasets.
 - 1 million instances
 - 30 features used in the fraud detection
 - "fraud_bool" = 1 for fraudulent case, =0 for legitimate case
- Fraud is highly imbalanced task
 - Focus on measure FPR, FNR, instead of accuracy
 - Too much non-fraud case, even if the model with non-predict power may cause high accuracy, measure accuracy meaningless
 - FPR: how often the model incorrectly flags non-fraud as fraud
 - Target: use FPR at a fixed TPR=0.8
 - What is the lowest FPR the model can achieve while catching 80% of all frauds?
 - Potential classifiers -> add to recommendation part, not included in the project codes
 - Subset of fraud
 - Boosting (weak classifier->stronger)

Analysis

- **Exploratory Data Analysis**
 - # of transactions (fraud) vs. (non-fraud)
 - Check missing values:
 - Missing values represented by (-1 or negative values) in the dataset → replace with NaN
 - 6 numeric columns with missing values
 - Deal with missing values
 - Features (Binary, Numerical, Categorical)
 - Numeric feature:
 - Numerical density distribution
 - Outliers, unbalanced datasets
 - Categorical feature:
 - Distribution of each categorical feature + fraud rates of each category for each feature
 - Demographical feature:
 - Income:
 - ◆ stacked bar chart, when the income is between 0.8 and 0.9, the number of fraudulent cases is significantly higher than other income values.
 - credit risk score:
 - ◆ Violin plot, the distribution of non fraudulent data is more significantly concentrated around credit_risk_score=100; fraudulent samples are more concentrated around higher credit_risk_score
 - customer age
 - ◆ Majority population (age 20-40), fraud mainly (age 30-50)
 - ◆ Age↑ fraud rate↑
- **Handle imbalanced data:**
 - NearMiss: Undersampling - Removes samples from the majority class
 - Cons: May lose information from the majority class
 - SMOTE: Oversampling - Creates synthetic samples for the minority class
 - Cons: Can overfit or create noisy synthetic data
 - ADASYN: Oversampling - Like SMOTE, but focuses more on harder-to-learn minority samples
 - Used when the classifier struggles with complex boundaries of minority class
 - Cons: Can overemphasize noisy minority class regions
- **Feature Engineering**
 - Test for Multicollinearity
 - Test mutual info (numerical features): # of info shared between two r.v. (nonlinear relation?)
- **ML Models → Feature Importance**
 - Data splits:
 - 80% train, 20% test
 - Column "Month": first six month as training, last two month (recent) for validation
 - Logistic regression (baseline)
 - QDA
 - ~~KNN~~
 - ~~LDA~~
 - Tree Model:
 - Decision Tree
 - Random Forest

- XGBoost

Paper

- Introduction & Background & Dataset
- Exploratory Data Analysis
- Data Preprocessing
- Modeling & Result Analysis
- Application & Future Improvement / Recommendations on data
- References

Submissions

1. 30-min presentation (Data)
2. Submissions: code, folder of data, report, presentation ppt